Pharmaceutical Report

Group 22

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Executive Summary

This project investigated how pharmaceutical dispensing varied across New Zealand health districts from 2020 to 2024, with a particular focus on two medications used in the treatment of dementia: donepezil hydrochloride and rivastigmine. Pharmaceutical dispensing data were combined with the New Zealand Index of Socioeconomic Deprivation and annual population estimates for people aged 65 years and over. The pharmaceutical data were restricted to dispensing records, district names were harmonised across datasets, and suppressed values reported as fewer than six were approximated as three where numerical values were required. Population-adjusted dispensing rates were also calculated to allow more meaningful comparisons between districts with different older population sizes.

Exploratory analysis showed that dispensing counts for both donepezil hydrochloride and rivastigmine increased nationally between 2020 and 2024, although neither medication was among the most widely dispensed pharmaceuticals overall. Dispensing also varied substantially between health districts. The size of the population aged 65 years and over was strongly related to raw dementia-medication dispensing counts, indicating that differences in older population size account for a considerable amount of the variation between districts and supporting the use of population-adjusted rates. In 2024, donepezil hydrochloride generally had substantially higher dispensing rates than rivastigmine, while the districts with the highest rates differed between the two medications.

The exploratory analysis also suggested that socioeconomic deprivation may be related to dementia-medication dispensing differently depending on the medication and year. For donepezil hydrochloride, the observed relationship between deprivation and dispensing rates became more positive up to 2022 before decreasing in later years, while rivastigmine showed increasingly negative trends across the study period. However, these patterns should be interpreted cautiously because the analysis includes only around 19 districts per year, the same districts are repeatedly observed, deprivation varies little over the study period, and uncertainty is particularly high for rivastigmine.

Overall, the results provide exploratory evidence of temporal, geographic and socioeconomic variation in dementia-medication dispensing across New Zealand. The findings describe associations rather than causal relationships and may also reflect differences in population structure, healthcare access, prescribing practices, diagnosis rates and other factors not measured in the available data.

Background

Medicines are an important part of healthcare, and examining pharmaceutical dispensing data can help us understand how medicine use varies across New Zealand, between geographic areas and over time. Dispensing patterns may be influenced by differences in population size, age structure, socioeconomic conditions and healthcare needs. This project uses pharmaceutical dispensing data to explore these patterns, with a particular focus on medicines associated with dementia.

Dementia describes a group of conditions that cause a progressive decline in cognitive functions such as memory, thinking and the ability to carry out everyday activities. It is more common among older people and can create substantial healthcare and support needs. Understanding patterns in dementia-related pharmaceutical dispensing is therefore important, particularly as differences may exist between geographic areas and population groups.

The group's analysis focuses on Donepezil hydrochloride and Rivastigmine, two medicines used in the management of symptoms associated with dementia. We examine how dispensing of these medicines changed between 2020 and 2024 and how patterns differed between New Zealand health districts. However, raw dispensing counts alone may not provide a fair comparison between districts. For example, an area with a larger population aged 65 years and over may naturally have more dementia-related medicine dispensings. Population and socioeconomic deprivation information were therefore incorporated to provide additional context when examining geographic differences.

The pharmaceutical data were obtained from Te Whatu Ora – Health New Zealand and contain aggregated information about medicines dispensed through community pharmacies in New Zealand. The initial exploratory analysis examines general pharmaceutical patterns, including commonly dispensed medicines, changes over time, geographic variation and therapeutic groups, before focusing more closely on Donepezil and Rivastigmine.

Data

Three main sources of data were used: pharmaceutical dispensing data, socioeconomic deprivation data, and population data. The pharmaceutical dispensing data were obtained from Health New Zealand | Te Whatu Ora's Pharmaceutical Data Web Tool. The socioeconomic deprivation data and population estimates were obtained from [https://explore.data.stats.govt.nz/vis?pg=0&bp=true&snb=19&tm=deprivation%20index&hc%5BNew%20Zealand%20index%20of%20socioeconomic%20deprivation%5D=&df%5Bds%5D=ds-nsiws-disseminate&df%5Bid%5D=CEN23_DEP_019&df%5Bag%5D=STATSNZ&df%5Bvs%5D=1.0&dq=2023.99999..9999.99.99&to%5BTIME%5D=false] 'Socioeconomic deprivation data'

[https://explore.data.stats.govt.nz/vis?pg=0&bp=true&snb=3&tm=Population%20estimates%20health%20district&hc%5BArea%5D=&df%5Bds%5D=ds-nsiws-disseminate&df%5Bid%5D=POPES_SUB_002&df%5Bag%5D=STATSNZ&df%5Bvs%5D=1.0&dq=2020%2B2021%2B2022%2B2023%2B2024%2B2025.3.4%2B14%2B15%2B16%2B17%2B18%2B19.100%2B101%2B102%2B103%2B200%2B201%2B202%2B203%2B204%2B300%2B301%2B302%2B303%2B304%2B400%2B401%2B402%2B403%2B404%2B900&to%5BTIME%5D=false&isAvailabilityDisabled=false]'Population Estimates Data'. These datasets have different levels of granularity and were integrated where required for the dementia analysis.

The pharmaceutical data were provided in several CSV files, including Data_ByChemical.csv, Data_ByTG2.csv, Datya_ByTG3.csv and PharmaceuticalsLookup.csv. The chemical-level dataset was the main dataset used for analysing individual medicines. Each row represents an aggregated combination of characteristics such as pharmaceutical chemical, health district, year and dispensing type, rather than an individual patient. The TG2 and TG3 datasets contain more aggregated therapeutic-group and subgroup information, while the lookup table connects individual pharmaceutical chemicals to their corresponding therapeutic classifications.

The socioeconomic deprivation data provide information on deprivation across geographic areas. For the analysis, these data were used to calculate a population-weighted deprivation measure for each health district, providing district-level socioeconomic context. The population data contain population estimates by geographic area, year and age group. For the dementia analysis, these were aggregated to obtain estimates of the population aged 65 years and over for each district and year. This allowed dementia-related dispensing patterns to be considered relative to the size of the older population rather than comparing districts using raw dispensing counts alone.

Variable	R data type	Description
Chemical	Character	Name of the pharmaceutical chemical
ChemID	Character	Identifier associated with the pharmaceutical

chemical

Variable	R data type	Description
District	Character	Health district or New Zealand national aggregate
YearDisp	Numerical	Year in which the dispensing was recorded
NumDisps	Numerical after cleaning	Number of pharmaceutical dispensings
NumPpl	Numerical after cleaning	Number of people associated with the dispensing record

Figure 1. Main variables in the chemical-level pharmaceutical dataset.

Although variables such as Chemical, District and Type are stored as character variables in R, they can be treated statistically as categorical variables. NumDisps and NumPpl were initially stored as character variables because some observations contain the value <6. After treatment of these suppressed observations, they were converted to numerical variables.

Data quality and preparation

Before the main analysis, the pharmaceutical data were checked for missing values, duplicate records and variable structure. Records were then filtered to include Type = "Dispensings", excluding initial dispensing records before suppression was assessed. After this filtering, 4127 records had suppressed NumDisps values and 10147 records had suppressed NumPpl values.

Suppression occurs when a small count is reported as <6 rather than providing its exact value. For numerical analysis, these values were replaced with 3 for both NumDisps and NumPpl, allowing the observations to remain in the analysis rather than being removed. However, the actual suppressed values are unknown, so imputing a value of 3 introduces a small amount of uncertainty into calculated totals, rates and subsequent analyses.

Integration of the three data sources also required attention to the join keys. District names were not completely consistent between datasets because of differences in naming conventions and the inclusion of Māori macrons. For example, Tairawhiti and Waitemata in the pharmaceutical data were standardised to Tairawhiti and Waitemata. Capital & Coast and Hutt Valley were also combined as Capital, Coast and Hutt Valley where required to match the geographic structure of the other data. Standardising these names was necessary because inconsistent join keys could result in records for the same geographic area failing to match.

After standardisation, pharmaceutical and deprivation information could be linked by district, while population information was linked using district and year. These integrated data provide the basis for examining whether differences in Donepezil and Rivastigmine dispensing across New Zealand are associated with geographic, demographic and socioeconomic characteristics.

Ethics, Privacy and Security

Ethical considerations

The project uses aggregated pharmaceutical dispensing data to investigate patterns involving donepezil hydrochloride, rivastigmine and other widely distributed medications across New Zealand health districts. Because the data is aggregated geographically and does not capture all factors that may influence medication dispensing, relationships observed in the analysis cannot be assumed to represent individual behaviour or direct causal relationships.

In particular, an observed relationship between socioeconomic deprivation and medication dispensing should not be interpreted as evidence that deprivation directly causes differences in medication use. Differences between districts may also reflect factors that are not included in the analysis, such as access to healthcare, prescribing practices, demographic composition, population size, or differences in the prevalence and diagnosis of particular conditions. For this reason, the analysis describes associations and trends rather than causal relationships.

Care is also required when discussing areas with relatively high levels of deprivation. Results should be presented in a way that avoids stigmatising particular districts or populations. A higher or lower dispensing rate does not automatically indicate better or worse healthcare because observed dispensing patterns may reflect several demographic, clinical and structural factors.

Māori data sovereignty is also relevant when analysing health information in Aotearoa New Zealand. Although this project uses aggregated data rather than identifiable Māori health records, analyses of geographic and socioeconomic health patterns may still contribute to conclusions about Māori communities. Results involving Māori populations should therefore be interpreted and communicated respectfully and within their wider social and historical context. With access to more detailed population-level or ethnicity-specific health information, the project would require greater consideration of Māori governance over the collection, interpretation, storage and use of Māori data.

Privacy considerations

The pharmaceutical dataset used in the project is aggregated at health-district level rather than individual-patient level. Individual-level medication records, names, NHI numbers and other direct patient identifiers were therefore neither required nor used in the analysis.

Small counts in the pharmaceutical dataset are represented as "<6" to reduce the risk of identifying patients in groups with very few observations. These suppressed observations were identified during data preparation. Where numerical values were required for analysis, suppressed counts were approximated as 3, the midpoint of the possible range from 1 to 5. This allows suppressed observations to remain in the analysis without claiming that their true confidential values are known.

This approximation introduces uncertainty, particularly for medications or districts with relatively small dispensing counts. Results affected by these observations should therefore be interpreted cautiously, and the imputed value of 3 should not be presented as the actual underlying count.

Suppression also does not completely eliminate disclosure risk. In some circumstances, suppressed values could potentially be inferred by comparing totals, subtotals, years, medication categories or other overlapping published aggregates. For example, if a total and most of its component counts were available, the remaining suppressed value could potentially be estimated through subtraction. We did not attempt to reverse-engineer suppressed observations; instead, suppressed values were consistently treated as unknown values within the range of 1 to 5 and approximated only where required for analysis. This illustrates why privacy must be considered across combinations of outputs rather than only within individual tables.

Security considerations

For this project, we worked only with the datasets required for the analysis. The pharmaceutical data was already aggregated and did not contain names, NHI numbers or other direct patient identifiers. The original pharmaceutical dataset was loaded without being overwritten, while cleaned and transformed versions were created as separate data objects and derived output files. This helps preserve the original source while allowing the analytical transformations to remain reproducible.

The analysis code also explicitly records important processing decisions, including the identification and treatment of suppressed values, the restriction of the main analysis to total dispensings, the harmonisation of district geographies and the creation of population-adjusted dispensing rates. Keeping these transformations in code allows them to be reviewed and reproduced rather than being performed manually.

We also avoided intentionally reconstructing suppressed values when producing analytical outputs. However, because suppressed values may sometimes be inferred by combining multiple aggregates, tables and visualisations should be reviewed before publication to ensure that combinations of reported results do not inadvertently reveal small confidential counts.

With greater resources, and particularly if individual-level or more detailed health data were used, stronger security controls would be required. These could include restricted user access, secure research environments, encrypted storage and transfer, formal approval processes, audit logs and systematic disclosure-risk assessment before releasing results. More detailed Māori health data would also require stronger consideration of Māori data sovereignty and appropriate Māori governance over access, storage, interpretation and publication.

Exploratory Data Analysis

The graph below shows an unequal distribution of dispensing of medications of the 15 most widely distributed chemicals. Paracetamol is by far the most dispensed, with over double the amount of dispensings compared to the aspirin, the chemical in 4th place. The second and third most widely distributed medications are atorvastatin and omeprazole. Overall, the distribution has a long tail. Donepezil hydrochloride and Rivastigmine aren't highly distributed enough to make the top 15, with the former ranking 186th across the 5 years of data, and the latter ranking 469th over the same period.

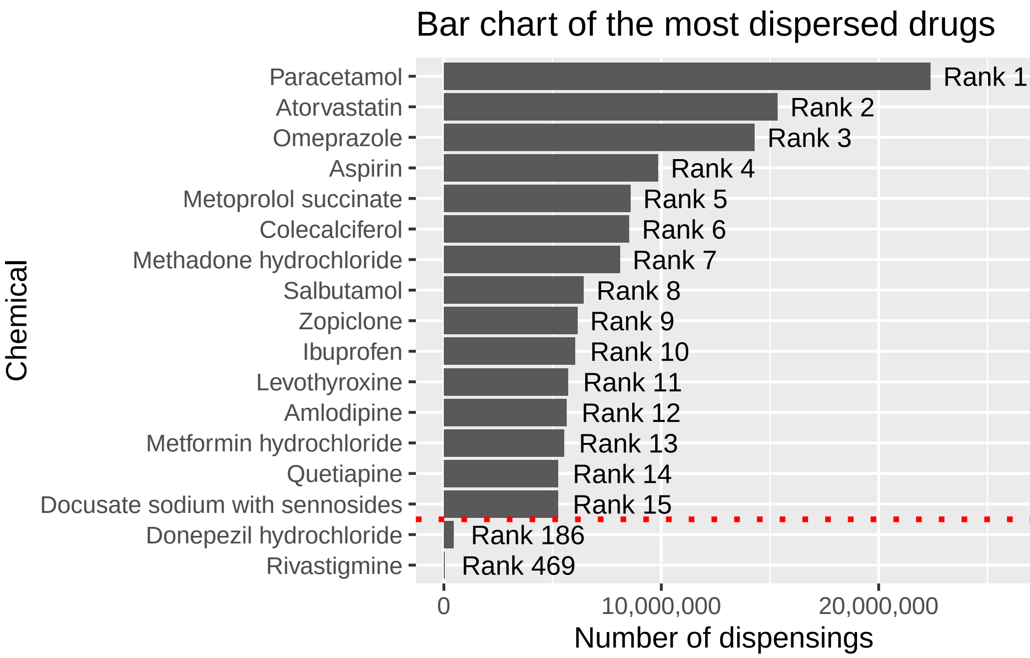

Figure 2. Bar chart representing the count of dispensing of the most widely dispensed drugs in comparison with the dementia medication.

Looking at the line charts below shows the dispensing count for the two primary dementia medications annually from 2020 to 2024. Both show an upward trend of distributions over time.

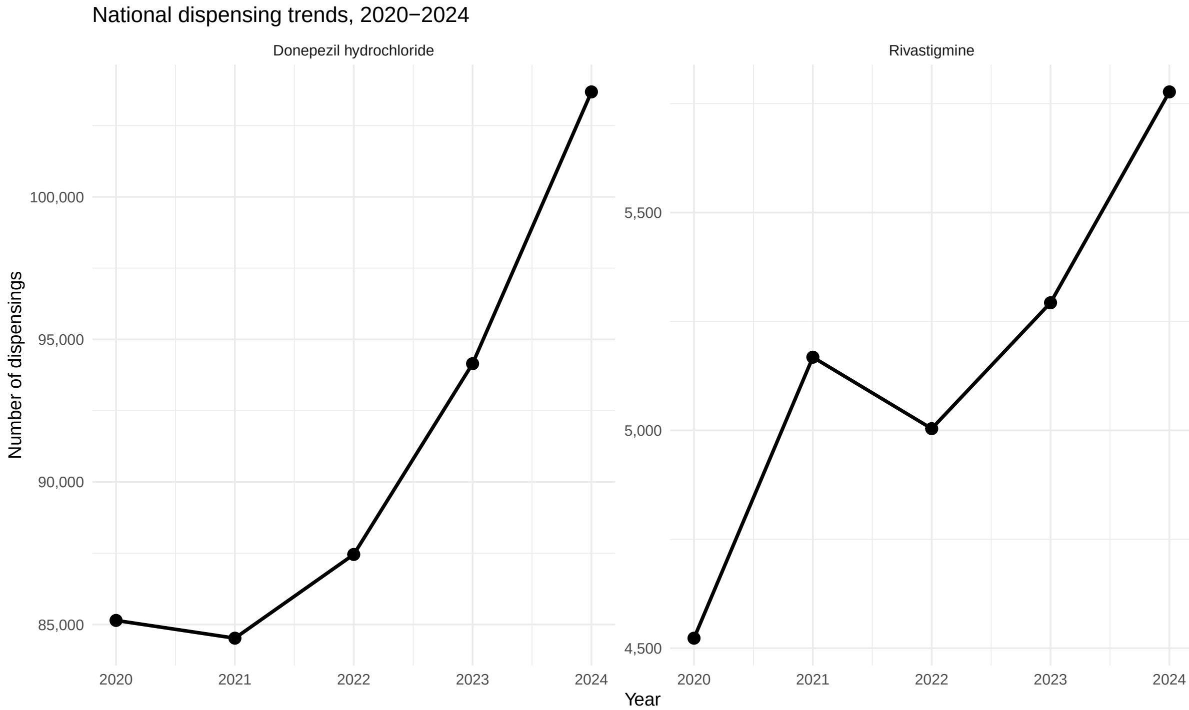

Figure 3. The change in dispensing of the dementia medication over time.

The table below shows the same information in table format, broken down by count and ranks of the two dementia medications of interest. There has been an upwards trend in dispensing for both of these medications, while rivastigmine has had its rank reduced in recent years. This is because while the dispensing of this medication has increased in these years, not as much as other medication that had a lower dispensing count that rivastigmine in 2021.

Year	DH count	R count	DH rank	R rank
2020	85142	4523	188	466
2021	84519	5168	189	454
2022	87456	5004	192	457
2023	94148	5293	183	467
2024	103683	5777	179	474

Figure 4. Count table of distribution of dementia medication, along with the distribution rank relative to other medications.

It is useful to know how frequently medications are distributed in a high concentration to a few health districts relative to an even spread across all health districts. One way to measure this is by using the proportion of dispensings going toward the health district where most of the dispensings occur. There are 19 health districts, and if the dispensings occurred equally across said districts, this metric would take the value of 1/19. In the opposite extreme, if the medication was distributed in only 1 district, the metric would take the value of 1. Below is a histogram showing how each of the medications fall on this proportion scale.

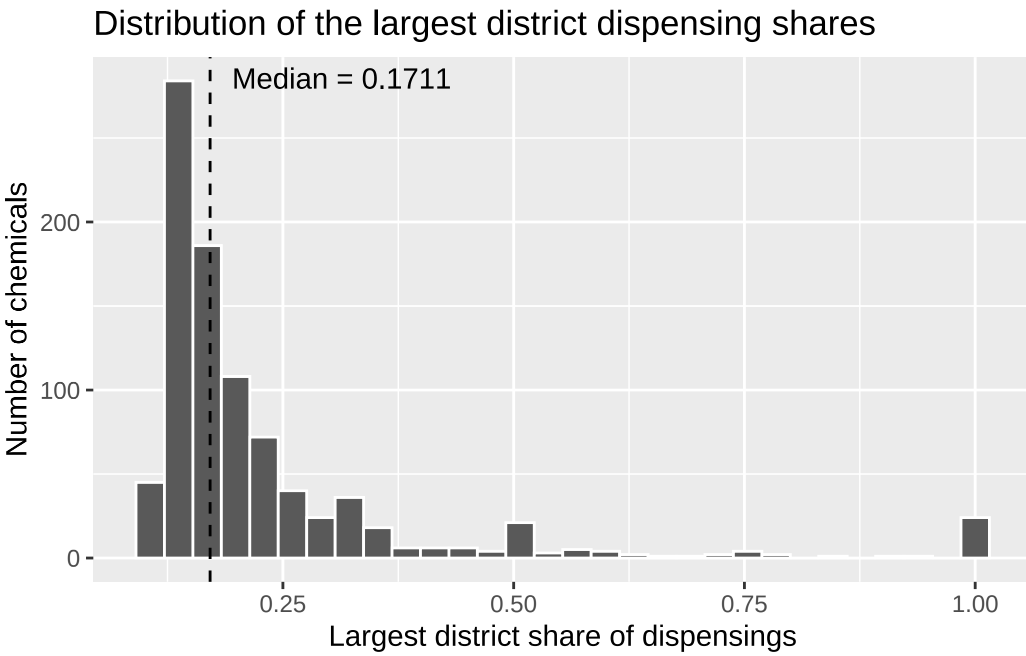

Figure 5. Histogram showing the disrtibution of the proportion of dispensings to the district with the highest concentration of dispensings.

The median value is 0.1711, meaning that a typical medication will have roughly 17% of their medication go towards the district that leads in dispensing. The lowest observed proportion is 0.106283 meaning that no medication is equally distributed across all districts. There's a small group of medications that have a proportion value of 1, meaning that they are only distributed in a single district. Note that this metric doesn't speak to the number of dispensings for these medications, and are likely very niche with an estimated mean of 7.38 and max of 55. The medications with these lower dispensings may be very niche, available from retailers outside of pharmacies or not recorded in some districts, but this is speculative.

Dementia as a risk is strongly associated with old age. This makes the districts with a higher count of aged population more relevant to this analysis than total population size in the health district. The graphs below show a positive relationship between aged population count and dispensings of the two key dementia medications. This relationship is stronger for Donepezil Hydrochloride, but both show that there are some observations above the line of best fit on the higher end of the district aged population. These observations belong to the Canterbury district, and as the years increase it can be seen diverging further away from the line of best fit. Overall, this high correlation means that the older population size accounts for a large amount of the inter-district variation.

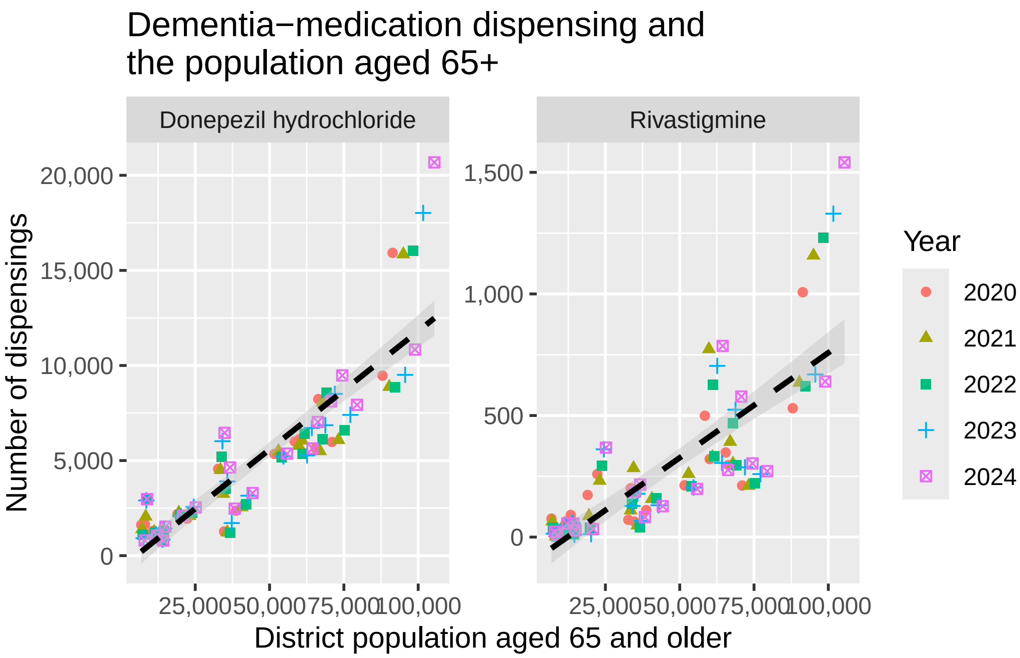

Figure 6. An annual breakdown of how dispensing counts vary by count of population aged 65+ from 2020 to 2024.

Crude rates were calculated as annual dispensings per 1,000 people aged 65 years and over for the 2024 year. The table below shows the variation in these dispensing rates across the different districts.

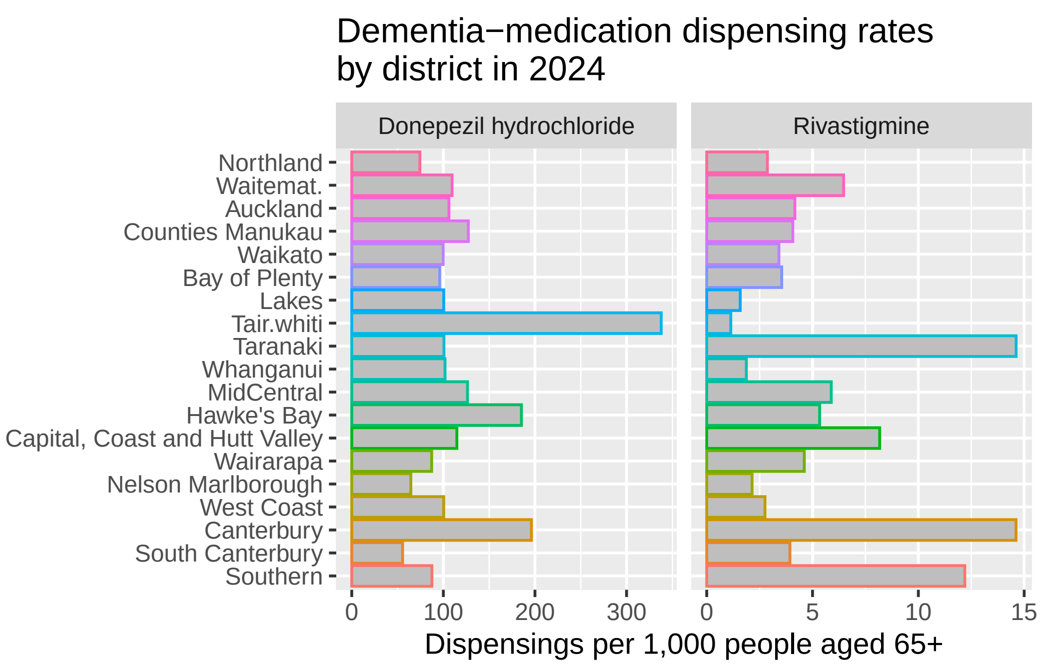

Figure 7. Bar chart displaying the dispensing rate per count of population aged 65+ broken down by district.

The first observation of note is that donepezil hydrochloride has a much higher rate than rivastigmine. Tairāwhiti has the highest overall rate when you take into account said larger rates for donepezil hydrochloride, as it is the district that has the highest rate for this medication. Canterbury and Hawke's Bay have the second and third highest rates respectively. Alternatively, Rivastigmine has the highest concentration of dispensing in Canterbury, Taranaki and Southern. Taranaki doesn't stand out for being high or low with donepezil hydrochloride, and Tairāwhiti has the lowest dispensing rate for rivastigmine. This could be related to the high proportion of the population of Tairāwhti falling higher on the latest (2023) NZ deprivation scale as seen below.

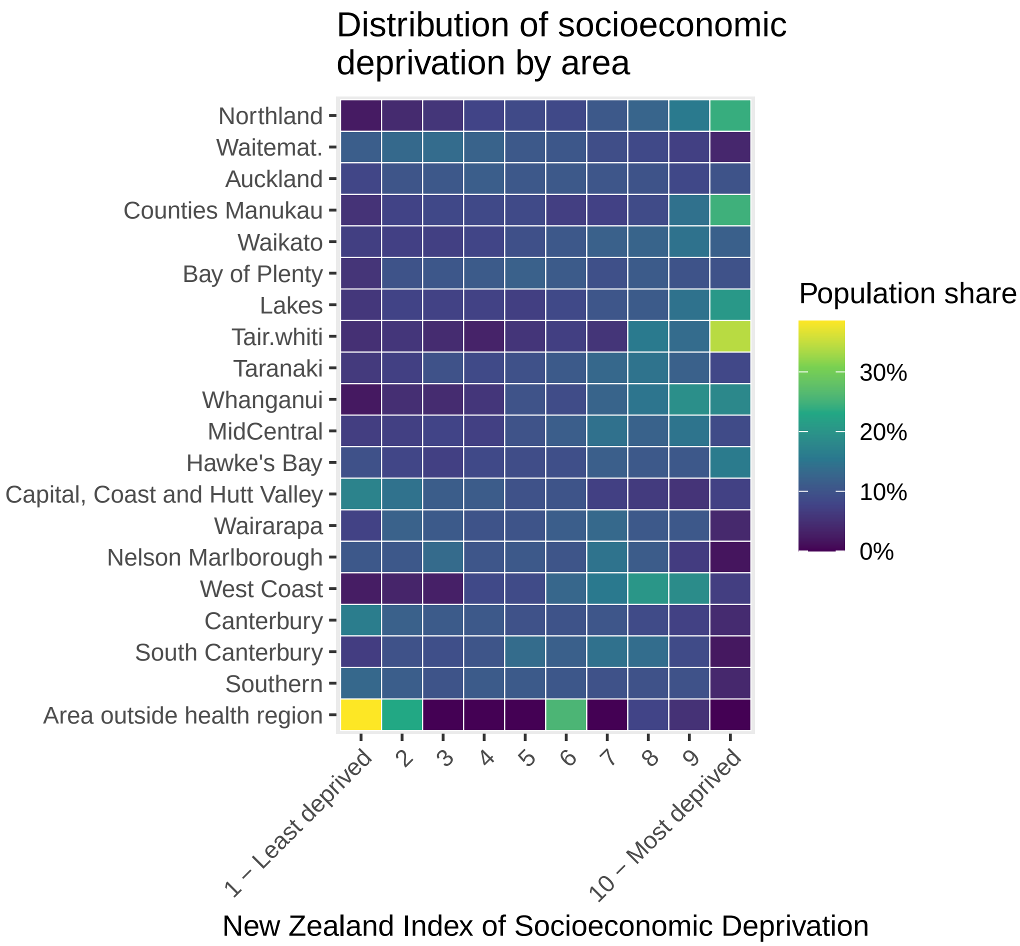

Figure 8. Heatmap showing the count of the district population falling into each deprivation index.

The plots below show how the average dispensing rate per aged person changes for the two dementia medications over the deprivation index. For donepezil hydrochloride, the trend in the relationship between the dispensing rate and the deprivation index increases as time increases, peaking at 2022, followed by a decreasing trend in subsequent years. Alternatively, these trends have consistently steeper downward trends as the years increase. This is a strong argument to test interaction effects between year and deprivation index when predicting the number of medication dispensings.

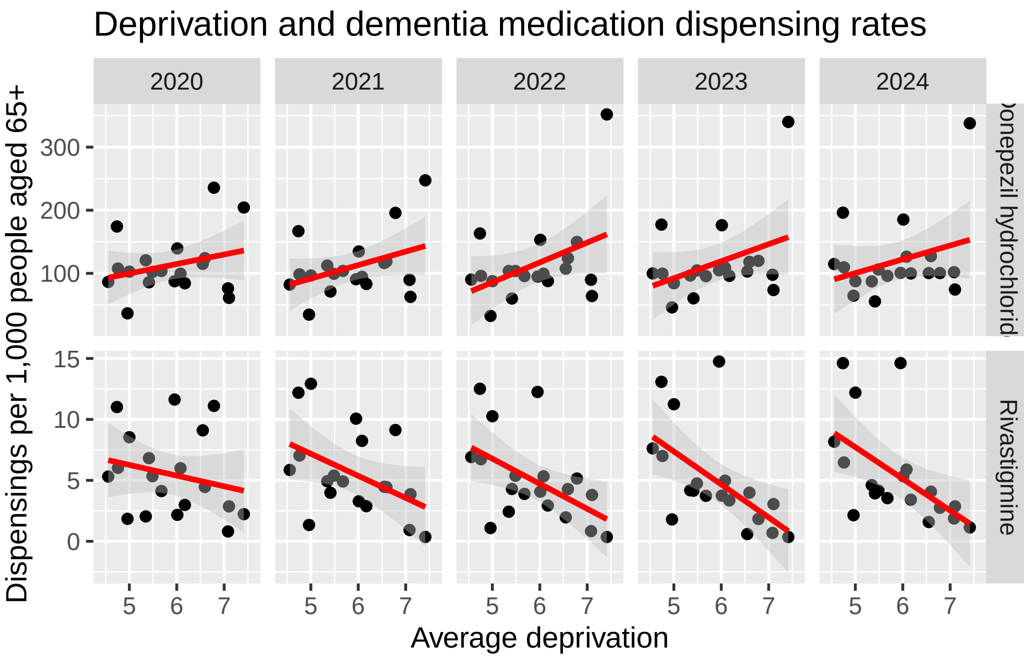

Figure 9. Scatterplot displaying the districts annual dispensing rate per population aged 65+ over average deprivation for dementia medication.

These results remain exploratory. There are only about 19 districts per year, the same districts are observed repeatedly, and district deprivation changes little over the study period. Confidence intervals are wide, especially for rivastigmine, and the slopes may be sensitive to influential districts such as Tairāwhiti, Canterbury or Taranaki.

Individual Contributions

Executive Summary:

Written by Amine.

Background and Data:

Written by Arnab.

Ethic, Privacy and Security:

Colton, Amine and Arnab all contributed 4 points, which were summarised by Amine.

EDA:

Colton, Amine and Arnab all did individual contributions toward the plots and commentary. Collated by Colton.